Proof of Theorem 4E.1: Since suspension is an isomorphism in any reduced cohomology theory, and the suspension of any CW complex is connected, it suffices to restrict attention to connected CW complexes. Each functor h^n satisfies the homotopy, wedge, and Mayer–Vietoris axioms, as we noted earlier, so the preceding theorem gives CW complexes K_n with $h^n(X) = \langle X, K_n\rangle$. It remains to show that the natural isomorphisms $h^n(X) \approx h^{n+1}(\Sigma X)$ correspond to weak homotopy equivalences $K_n \to \Omega K_{n+1}$. The natural isomorphism $h^n(X) \approx h^{n+1}(\Sigma X)$ corresponds to a natural bijection $\langle X, K_n\rangle \approx \langle \Sigma X, K_{n+1}\rangle = \langle X, \Omega K_{n+1}\rangle$ which we call Φ. The naturality of this bijection gives, for any map $f: X \to K_n$, a commutative diagram as at the right. Let $\varepsilon_n = \Phi(\mathbb{1}): K_n \to \Omega K_{n+1}$. Then using commutativity we have $\Phi(f) = \Phi f^*(\mathbb{1}) = f^*\Phi(\mathbb{1}) = f^*(\varepsilon_n) = \varepsilon_n f$, which says that the map $\Phi: \langle X, K_n\rangle \to \langle X, \Omega K_{n+1}\rangle$ is composition with ε_n. Since Φ is a bijection, if we take X to be S^i, we see that ε_n induces an isomorphism on π_i for all i, so ε_n is a weak homotopy equivalence and we have an Ω-spectrum.

$$\begin{array}{ccc} \langle K_n, K_n\rangle & \xrightarrow{f^*} & \langle X, K_n\rangle \\ \downarrow{\scriptstyle\Phi} & & \downarrow{\scriptstyle\Phi} \\ \langle K_n, \Omega K_{n+1}\rangle & \xrightarrow{f^*} & \langle X, \Omega K_{n+1}\rangle \end{array}$$

There is one final thing to verify, that the bijection $h^n(X) = \langle X, K_n\rangle$ is a group isomorphism, where $\langle X, K_n\rangle$ has the group structure that comes from identifying it with $\langle X, \Omega K_{n+1}\rangle = \langle \Sigma X, K_{n+1}\rangle$. Via the natural isomorphism $h^n(X) \approx h^{n+1}(\Sigma X)$ this is equivalent to showing the bijection $h^{n+1}(\Sigma X) = \langle \Sigma X, K_{n+1}\rangle$ preserves group structure. For maps $f, g: \Sigma X \to K$, the relation $T_u(f+g) = T_u(f) + T_u(g)$ means $(f+g)^*(u) = f^*(u) + g^*(u)$, and this holds since $(f+g)^* = f^* + g^*: h(K) \to h(\Sigma X)$ by Lemma 4.60. $\square$

4.F Spectra and Homology Theories

We have seen in §4.3 and the preceding section how cohomology theories have a homotopy-theoretic interpretation in terms of Ω-spectra, and it is natural to look for a corresponding description of homology theories. In this case we do not already have a homotopy-theoretic description of ordinary homology to serve as a starting point. But there is another homology theory we have encountered which does have a very homotopy-theoretic flavor:

Proposition 4F.1. *Stable homotopy groups $\pi_n^s(X)$ define a reduced homology theory on the category of basepointed CW complexes and basepoint-preserving maps.*

Proof: In the preceding section we reformulated the axioms for a cohomology theory so that the exactness axiom asserts just the exactness of $h^n(X/A) \to h^n(X) \to h^n(A)$ for CW pairs (X, A). In order to derive long exact sequences, the reformulated axioms require also that natural suspension isomorphisms $h^n(X) \approx h^{n+1}(\Sigma X)$ be specified as part of the cohomology theory. The analogous reformulation of the axioms for a homology theory is valid as well, by the same argument, and we shall use this in what follows.

For stable homotopy groups, suspension isomorphisms $\pi_n^s(X) \approx \pi_{n+1}^s(\Sigma X)$ are automatic, so it remains to verify the three axioms. The homotopy axiom is apparent. The exactness of a sequence $\pi_n^s(A) \to \pi_n^s(X) \to \pi_n^s(X/A)$ follows from exactness of $\pi_n(A) \to \pi_n(X) \to \pi_n(X,A)$ together with the isomorphism $\pi_n(X,A) \approx \pi_n(X/A)$ which holds under connectivity assumptions that are achieved after sufficiently many suspensions. The wedge sum axiom $\pi_n^s(\bigvee_\alpha X_\alpha) \approx \bigoplus_\alpha \pi_n^s(X_\alpha)$ reduces to the case of finitely many summands by the usual compactness argument, and the case of finitely many summands reduces to the case of two summands by induction. Then we have isomorphisms $\pi_{n+i}(\Sigma^i X \vee \Sigma^i Y) \approx \pi_{n+i}(\Sigma^i X \times \Sigma^i Y) \approx \pi_{n+i}(\Sigma^i X) \bigoplus \pi_{n+i}(\Sigma^i Y)$, the first of these isomorphisms holding when $n+i < 2i-1$, or $i > n+1$, since $\Sigma^i X \vee \Sigma^i Y$ is the $(2i-1)$-skeleton of $\Sigma^i X \times \Sigma^i Y$. Passing to the limit over increasing i, we get the desired isomorphism $\pi_n^s(X \vee Y) \approx \pi_n^s(X) \bigoplus \pi_n^s(Y)$. $\square$

A modest generalization of this homology theory can be obtained by defining $h_n(X) = \pi_n^s(X \wedge K)$ for a fixed complex K. Verifying the homology axioms reduces to the case of stable homotopy groups themselves by basic properties of smash product:

- $h_n(X) \approx h_{n+1}(\Sigma X)$ since $\Sigma(X \wedge K) = (\Sigma X) \wedge K$, both spaces being $S^1 \wedge X \wedge K$.
- The exactness axiom holds since $(X \wedge K)/(A \wedge K) = (X/A) \wedge K$, both spaces being quotients of $X \times K$ with $A \times K \cup X \times \{k_0\}$ collapsed to a point.
- The wedge axiom follows from distributivity: $(\bigvee_\alpha X_\alpha) \wedge K = \bigvee_\alpha (X_\alpha \wedge K)$.

The coefficients of this homology theory are $h_n(S^0) = \pi_n^s(S^0 \wedge K) = \pi_n^s(K)$. Suppose for example that K is an Eilenberg–MacLane space $K(G,n)$. Because $K(G,n)$ is $(n-1)$-connected, its stable homotopy groups are the same as its unstable homotopy groups below dimension $2n$. Thus if we shift dimensions by defining $h_i(X) = \pi_{i+n}^s(X \wedge K(G,n))$ we obtain a homology theory whose coefficient groups below dimension n are the same as ordinary homology with coefficients in G. It follows as in Theorem 4.59 that this homology theory agrees with ordinary homology for CW complexes of dimension less than $n-1$.

This dimension restriction could be removed if there were a 'stable Eilenberg–MacLane space' whose stable homotopy groups were zero except in one dimension. However, this is a lot to ask for, so instead one seeks to form a limit of the groups $\pi_{i+n}^s(X \wedge K(G,n))$ as n goes to infinity. The spaces $K(G,n)$ for varying n are related by weak homotopy equivalences $K(G,n) \to \Omega K(G,n+1)$. Since suspension plays such a large role in the current discussion, let us consider instead the corresponding map $\Sigma K(G,n) \to K(G,n+1)$, or to write this more concisely, $\Sigma K_n \to K_{n+1}$ This induces a map $\pi_{i+n}^s(X \wedge K_n) = \pi_{i+n+1}^s(X \wedge \Sigma K_n) \to \pi_{i+n+1}^s(X \wedge K_{n+1})$. Via these maps, it then makes sense to consider the direct limit as n goes to infinity, the group $h_i(X) = \varinjlim \pi^s_{i+n}(X \wedge K_n)$. This gives a homology theory since direct limits preserve exact sequences so the exactness axiom holds, and direct limits preserve isomorphisms so the suspension isomorphism and the wedge axiom hold. The coefficient groups of this homology theory are the same as for ordinary homology with G coefficients since $h_i(S^0) = \varinjlim \pi^s_{i+n}(K_n)$ is zero unless $i = 0$, when it is G. Hence this homology theory coincides with ordinary homology by Theorem 4.59.

To place this result in its natural generality, define a **spectrum** to be a sequence of CW complexes K_n together with basepoint-preserving maps $\Sigma K_n \to K_{n+1}$. This generalizes the notion of an Ω-spectrum, where the maps $\Sigma K_n \to K_{n+1}$ come from weak homotopy equivalences $K_n \to \Omega K_{n+1}$. Another obvious family of examples is **suspension spectra**, where one starts with an arbitrary CW complex X and defines $K_n = \Sigma^n X$ with $\Sigma K_n \to K_{n+1}$ the identity map.

The homotopy groups of a spectrum K are defined to be $\pi_i(K) = \varinjlim \pi_{i+n}(K_n)$ where the direct limit is computed using the compositions

$$\pi_{i+n}(K_n) \xrightarrow{\Sigma} \pi_{i+n+1}(\Sigma K_n) \longrightarrow \pi_{i+n+1}(K_{n+1})$$

with the latter map induced by the given map $\Sigma K_n \to K_{n+1}$. Thus in the case of the suspension spectrum of a space X, the homotopy groups of the spectrum are the same as the stable homotopy groups of X. For a general spectrum K we could also describe $\pi_i(K)$ as $\varinjlim \pi^s_{i+n}(K_n)$ since the composition $\pi_{i+n}(K_n) \to \pi_{i+n+j}(K_{n+j})$ factors through $\pi_{i+n+j}(\Sigma^j K_n)$. So the homotopy groups of a spectrum are 'stable homotopy groups' essentially by definition.

Returning now to the context of homology theories, if we are given a spectrum K and a CW complex X, then we have a spectrum $X \wedge K$ with $(X \wedge K)_n = X \wedge K_n$, using the obvious maps $\Sigma(X \wedge K_n) = X \wedge \Sigma K_n \to X \wedge K_{n+1}$. The groups $\pi_i(X \wedge K)$ are the groups $\varinjlim \pi^s_{i+n}(K_n)$ considered earlier in the case of an Eilenberg–MacLane spectrum, and the arguments given there show:

Proposition 4F.2. *For a spectrum K, the groups $h_i(X) = \pi_i(X \wedge K)$ form a reduced homology theory. When K is the Eilenberg–MacLane spectrum with $K_n = K(G,n)$, this homology theory is ordinary homology, so $\pi_i(X \wedge K) \approx \widetilde{H}_i(X;G)$.* □

If one wanted to associate a cohomology theory to an arbitrary spectrum K, one's first inclination would be to set $h^i(X) = \varinjlim \langle \Sigma^n X, K_{n+i}\rangle$, the direct limit with respect to the compositions

$$\langle \Sigma^n X, K_{n+i}\rangle \xrightarrow{\Sigma} \langle \Sigma^{n+1} X, \Sigma K_{n+i}\rangle \longrightarrow \langle \Sigma^{n+1} X, K_{n+i+1}\rangle$$

For example, in the case of the sphere spectrum $S = \{S^n\}$ this definition yields the **stable cohomotopy groups** $\pi^i_s(X) = \varinjlim \langle \Sigma^n X, S^{n+i}\rangle$. Unfortunately the definition $h^i(X) = \varinjlim \langle \Sigma^n X, K_{n+i}\rangle$ runs into problems with the wedge sum axiom since the direct limit of a product need not equal the product of the direct limits. For finite wedge sums there is no difficulty, so we do have a cohomology theory for finite CW complexes. But for general CW complexes a different definition is needed. The simplest thing to do is to associate to each spectrum K an Ω-spectrum K' and let $h^n(X) = \langle X, K'_n \rangle$. We obtain K' from K by setting $K'_n = \varinjlim \Omega^i K_{n+i}$, the mapping telescope of the sequence $K_n \to \Omega K_{n+1} \to \Omega^2 K_{n+2} \to \cdots$. The Ω-spectrum structure is given by equivalences

$$K'_n = \varinjlim \Omega^i K_{n+i} \simeq \varinjlim \Omega^{i+1} K_{n+i+1} \xrightarrow{\kappa} \Omega \varinjlim \Omega^i K_{n+i+1} = \Omega K'_{n+1}$$

The first homotopy equivalence comes from deleting the first term of the sequence $K_n \to \Omega K_{n+1} \to \Omega^2 K_{n+2} \to \cdots$, which has negligible effect on the mapping telescope. The next map κ is a special case of the natural weak equivalence $\varinjlim \Omega Z_n \to \Omega \varinjlim Z_n$ that holds for any sequence $Z_1 \to Z_2 \to \cdots$. Strictly speaking, we should let K'_n be a CW approximation to the mapping telescope $\varinjlim \Omega^i K_{n+i}$ in order to obtain a spectrum consisting of CW complexes, in accordance with our definition of a spectrum.

In case one starts with a suspension spectrum $K_n = \Sigma^n K$ it is not necessary to take mapping telescopes since one can just set $K'_n = \bigcup_i \Omega^i \Sigma^{i+n} K = \bigcup_i \Omega^i \Sigma^i K_n$, the union with respect to the natural inclusions $\Omega^i \Sigma^i K_n \subset \Omega^{i+1} \Sigma^{i+1} K_n$. The union $\bigcup_i \Omega^i \Sigma^i X$ is usually abbreviated to $\Omega^\infty \Sigma^\infty X$. Another common notation for this union is QX. Thus $\pi_i(QX) = \pi_i^s(X)$, so Q is a functor converting stable homotopy groups into ordinary homotopy groups.

It follows routinely from the definitions that the homology theory defined by a spectrum is the same as the homology theory defined by the associated Ω-spectrum. One may ask whether every homology theory is defined by a spectrum, as we showed for cohomology. The answer is yes if one replaces the wedge axiom by a stronger **direct limit axiom**: $h_i(X) = \varinjlim h_i(X_\alpha)$, the direct limit over the finite subcomplexes X_α of X. The homology theory defined by a spectrum satisfies this axiom, and the converse is proved in [Adams 1971].

Spectra have become the preferred language for describing many stable phenomena in algebraic topology. The increased flexibility of spectra is not without its price, however, since a number of concepts that are elementary for spaces become quite a bit more subtle for spectra, such as the proper definition of a map between spectra, or the smash product of two spectra. For the reader who wants to learn more about this language a good starting point is [Adams 1974].

Exercises

1. Assuming the first two axioms for a homology theory on the CW category, show that the direct limit axiom implies the wedge sum axiom. Show that the converse also holds for countable CW complexes.

2. For CW complexes X and Y consider the suspension sequence

$$\langle X, Y \rangle \xrightarrow{\Sigma} \langle \Sigma X, \Sigma Y \rangle \xrightarrow{\Sigma} \langle \Sigma^2 X, \Sigma^2 Y \rangle \longrightarrow \cdots$$